

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397591
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Onizuka; Nobuyuki

Tire

Abstract

The tire **1** according to this disclosure is a tire having a carcass **3** comprised of at least one carcass ply **3a** extending between a pair of bead portions **13** through a pair of sidewall portions **12**. The tire **1** comprises: a conductive member **21** embedded in a tire side portion **14** comprised of the sidewall portion **12** and the bead portion **13**, either outside or inside the carcass **3** in the tire width direction, and extending over at least partial area in the tire circumferential direction and at least partial area in the tire radial direction in the tire side portion **14**; and at least one IC chip **22** embedded in the tire side portion **14**, and at least one of the at least one IC chip **22** is a contact IC chip **22c** that contacts the conductive member without a mechanical connection.

Inventors:	Onizuka; Nobuyuki (Tokyo, JP)
Applicant:	BRIDGESTONE CORPORATION (Tokyo, JP)
Family ID:	1000008782717
Assignee:	BRIDGESTONE CORPORATION (Tokyo, JP)
Appl. No.:	18/551540
Filed (or PCT Filed):	November 24, 2021
PCT No.:	PCT/JP2021/043055
PCT Pub. No.:	WO2023/276188
PCT Pub. Date:	January 05, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240166001 A1	May. 23, 2024

Foreign Application Priority Data

JP 2021-109729 Jun. 30, 2021

Publication Classification

Int. Cl.: B60C19/00 (20060101)

U.S. Cl.:

CPC B60C19/00 (20130101); B60C2019/004 (20130101)

Field of Classification Search

CPC: B60C (19/00); B60C (2019/004); B60C (23/0444); B60C (23/0452); H01Q (1/2241)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5218861	12/1992	Brown et al.	N/A	N/A
5388627	12/1994	Nakada	152/555	B60C 13/02
6228929	12/2000	Larson	524/495	C08L 23/22
6302173	12/2000	Mizuno	152/152.1	B60C 1/0016
6546982	12/2002	Brown	29/601	B60C 23/0433
11007826	12/2020	Nagayoshi	N/A	N/A
11981166	12/2023	Destraes et al.	N/A	N/A
2002/0148545	12/2001	Nanni	524/495	B60C 1/0016
2007/0256485	12/2006	Rensel	340/447	B60C 23/0493
2011/0095870	12/2010	Fenkany	340/10.3	H04Q 9/00
2012/0291936	12/2011	Lionetti et al.	N/A	N/A
2019/0366780	12/2018	Miklic et al.	N/A	N/A
2020/0108671	12/2019	Hosomi	N/A	N/A
2021/0001671	12/2020	Cukierman et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2275286	12/2010	EP	N/A
3237237	12/2016	EP	N/A
3237237	12/2018	EP	B60C 11/24
3578395	12/2018	EP	N/A
2004082775	12/2003	JP	N/A

2012240680	12/2011	JP	N/A
2016539047	12/2015	JP	N/A
2017132291	12/2016	JP	N/A
2019137150	12/2018	JP	N/A
2020055452	12/2019	JP	N/A
2020055453	12/2019	JP	N/A
2021514891	12/2020	JP	N/A
WO-2019076500	12/2018	WO	B29D 30/0061
2019175509	12/2018	WO	N/A
2021058905	12/2020	WO	N/A

OTHER PUBLICATIONS

English machine translation of WO-2019076500-A1. (Year: 2019). cited by examiner
Dec. 14, 2023, International Preliminary Report on Patentability issued in the International Patent Application No. PCT/JP2021/043055. cited by applicant
Jan. 11, 2022, International Search Report issued in the International Patent Application No. PCT/JP2021/043055. cited by applicant
Jul. 16, 2024, search result of the Extended European Search Report issued by the European Patent Office in the corresponding European Patent Application No. 21948491.2. cited by applicant

Primary Examiner: Dye; Robert C

Attorney, Agent or Firm: KENJA IP LAW PC

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to a tire.

BACKGROUND

(2) The configuration in which a communication device such as an RF tag is embedded in a tire has been known. For example, the PTL 1 discloses a tire with an RF tag embedded within the tire side portion.

CITATION LIST

Patent Literature

(3) PTL 1: JP 2016-539047 A

SUMMARY

Technical Problem

(4) However, in general, if an RF tag, in which the IC chip and antenna are mechanically connected by solder or other means, is embedded in the tire as it is, there is a risk of durability degradation due to wire breakage, etc. at such connection. For reinforcement, an RF tag could be coated with rubber or resin in advance, however this would complicate the structure and require advanced technology.

(5) Therefore, it is an object of this disclosure to provide a tire which can improve the durability of the communication device with a simple configuration.

Solution to Problem

(6) The tire according to this disclosure is, a tire having a carcass comprised of at least one carcass ply extending between a pair of bead portions through a pair of sidewall portions, wherein the tire

comprises: a conductive member embedded in a tire side portion comprised of the sidewall portion and the bead portion, either outside or inside the carcass in the tire width direction, and extending over at least partial area in the tire circumferential direction and at least partial area in the tire radial direction in the tire side portion; and at least one IC chip embedded in the tire side portion, wherein at least one of the at least one IC chip is a contact IC chip that contacts the conductive member without a mechanical connection.

Advantageous Effect

(7) According to this disclosure, it is possible to provide a tire which can improve the durability of the communication device with a simple configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings:

(2) FIG. 1 is a side view of the tire side portion of the tire according to the first embodiment of this disclosure, viewed from the outside in the tire width direction;

(3) FIG. 2 is a cross-sectional view in the tire width direction, illustrating a portion of the tire of FIG. 1 in cross-section along the A-A line in FIG. 1, together with a partially enlarged view; and

(4) FIG. 3 is a cross-sectional view in the tire width direction, illustrating a portion of the tire according to the second embodiment of this disclosure by the similar cross-section along the A-A line in FIG. 1.

DETAILED DESCRIPTION

(5) The tire according to this disclosure can be suitably used for any type of pneumatic tire, for example, the pneumatic tire for passenger cars, trucks, and buses, etc.

(6) The following is an illustrative description of the embodiments of the tire according to this disclosure with reference to the drawings.

(7) In each figure, common components and parts are marked with the same reference numerals/signs. In some drawings, the tire width direction is indicated by the sign “WD”, the tire radial direction is indicated by the sign “RD”, and the tire circumferential direction is indicated by the sign “CD”. In this document, the side closer to the tire lumen is referred to as the “inside the tire” and the side farther from the tire lumen is referred to as the “outside the tire”.

(8) FIGS. 1 and 2 illustrate a tire 1 according to the first embodiment of this disclosure. FIG. 1 is a side view of the tire side portion of the tire according to the first embodiment of this disclosure, viewed from the outside in the tire width direction. FIG. 2 is a cross-sectional view in the tire width direction, illustrating a portion (specifically, the portion on one side relative to the tire equatorial plane CL) of the tire of FIG. 1 in cross-section along the A-A line in FIG. 1, together with a partially enlarged view. FIG. 3 is a cross-sectional view in the tire width direction, illustrating a portion of the tire 10 according to the second embodiment of this disclosure by the similar cross-section along the A-A line in FIG. 1.

(9) The tire 1 and the tire 10 according to embodiments of this disclosure may be configured as any type of tire.

(10) Hereafter, unless otherwise noted, the positional relationship and dimensions of each element shall be measured under the reference condition, with the tire mounted on the applicable rim, filled with the prescribed internal pressure, and unloaded. When the tire is mounted on the applicable rim, filled with the prescribed internal pressure, and loaded with the maximum load, the width of the contact patch in contact with the road surface in the tire width direction is referred to as the ground contact width of the tire, and the edge of the ground contact patch in the tire width direction is referred to as the ground contact edge.

(11) As used herein, the term “applicable rim” refers to the standard rim in the applicable size

(Measuring Rim in ETRTO's STANDARDS MANUAL and Design Rim in TRA's YEAR BOOK) as described or as may be described in the future in the industrial standard, which is valid for the region in which the tire is produced and used, such as JATMA YEAR BOOK of JATMA (Japan Automobile Tyre Manufacturers Association) in Japan, STANDARDS MANUAL of ETRTO (The European Tyre and Rim Technical Organization) in Europe, and YEAR BOOK of TRA (The Tire and Rim Association, Inc.) in the United States. For sizes not listed in these industrial standards, the term “applicable rim” refers to a rim with a width corresponding to the bead width of the pneumatic tire. The term “applicable rim” includes current sizes as well as future sizes to be listed in the aforementioned industrial standards. An example of the “size as described in the future” could be the sizes listed as “FUTURE DEVELOPMENTS” in the ETRTO 2013 edition.

(12) As used herein, the term “prescribed internal pressure” refers to the air pressure (maximum air pressure) corresponding to the maximum load capacity of a single wheel in the applicable size and ply rating, as described in the aforementioned JATMA YEAR BOOK and other industrial standards. For sizes not listed in the aforementioned industrial standards, the term “prescribed internal pressure” refers to the air pressure (maximum air pressure) corresponding to the maximum load capacity specified for each vehicle in which the tire is mounted. Further, as used herein, the term “maximum load” means the load corresponding to the maximum load capacity in the tire of the applicable size described in the aforementioned industrial standards, or, for sizes not listed in the aforementioned industrial standards, the load corresponding to the maximum load capacity specified for each vehicle in which the tire is mounted.

First Embodiment

(13) As illustrated in FIG. 2, the tire **1** according to the first embodiment of this disclosure comprises a tread portion **11**, a pair of sidewall portions **12** extending inwardly in the tire radial direction from both ends of the tread portion **11** in the tire width direction, and a pair of bead portions **13** provided at the inner end of each sidewall portion **12** in the tire radial direction. The tread portion **11** is the portion of the tire **1** that extends in the tire width direction between a pair of the ground contact edges. The bead portion **13** is configured to be in contact with the rim when the tire **1** is mounted on the rim, both at the inside in the tire radial direction and outside in the tire width direction.

(14) The tire **1** has a pair of tire side portions **14** extending inwardly in the tire radial direction from both ends of the tread portion **11** in the tire width direction. The tire side portion **14** comprised of the sidewall portion **12** and the bead portion **13**.

(15) Further, the tire **1** also comprises a pair of bead cores **2a**, a pair of bead fillers **2b**, a carcass **3**, a belt **4**, a tread rubber **5**, a side rubber **6a**, and an inner liner **7**. The aforementioned sidewall portion **12** is at least the portion extending inwardly in the tire radial direction from the belt **4** and extending outwardly in the tire radial direction from the bead portion **13**.

(16) Each of the bead cores **2a** is embedded in the corresponding bead portion **13**. The bead core **2a** comprises a plurality of bead wires with rubber coating around them. However, the bead core **2a** may comprise a single bead wire. The bead wire is suitably composed of metal (e.g., steel). The bead wire can, for example, be made of monofilament or stranded wire. The bead wire may be composed of organic or carbon fibers.

(17) Each of the bead fillers **2b** is positioned outside relative to the corresponding bead core **2a** in the tire radial direction. The bead filler **2b** extends tapering outward in the tire radial direction. The bead filler **2b** is made of rubber, for example.

(18) The bead filler is sometimes referred to as “stiffener”.

(19) For example, when the tire **1** is configured as a pneumatic tire for a truck/bus, although not illustrated, the bead filler **2b** may comprises a plurality of (e.g., two) bead filler sections. The plurality of bead filler sections can differ in hardness from each other, for example. The plurality of bead filler sections are arranged (stacked) along the tire radial direction, for example.

(20) The carcass **3** straddles between a pair of bead cores **2a** and extends toroidally through a pair

of sidewall portions **12** and a tread portion **11**. In other words, the carcass **3** extends between a pair of bead portions **13** through a pair of sidewall portions **12**. The carcass **3** is comprised of at least one carcass ply **3a** (one ply in the example in FIG. 2). Each carcass ply **3a** includes one or more carcass cords and coating rubber covering the carcass cords. The carcass cord can be formed, for example, with monofilament or stranded wire.

(21) The carcass cord may be composed of organic fibers made of polyester, nylon, rayon, aramid, etc., or may be composed of metal (e.g., steel). When the tire **1** is configured as a pneumatic tire for a truck/bus, the carcass cord is suitably composed of metal (e.g., steel). When the tire **1** is configured as a passenger vehicle pneumatic tire, the carcass cord is suitably composed of organic fibers made of polyester, nylon, rayon, aramid, or the like.

(22) The carcass ply **3a** comprises a ply body portion **3M** positioned between a pair of bead cores **2a**. The carcass ply **3a** may further comprise ply turn-up portions **3T**, which are folded from both ends of the ply body portion **3M** around the bead cores **2a** from the inside to the outside in the tire width direction. However, the carcass ply **3a** does not have to comprise the ply turn-up portions **3T**. The carcass **3** is suitably of a radial structure, but may also be of a bias structure.

(23) The belt **4** is disposed outside the crown portion of the carcass **3** in the tire radial direction. The belt **4** comprises at least one (two in the example in FIG. 2) belt layers **4a**. Each belt layer **4a** includes one or more belt cords and coating rubber covering the belt cords. The belt cord can be formed, for example, with monofilament or stranded wire. The belt cord may be composed of metal (e.g., steel) or of organic fibers comprising polyester, nylon, rayon, aramid, etc.

(24) The tread rubber **5** is positioned in the tread portion **11**, outside the belt **4** in the tire radial direction. The tread rubber **5** constitutes the tread surface, which is the outer surface of the tread portion **11** in the tire radial direction. The tread pattern is formed on the tread surface.

(25) The side rubber **6a** is positioned in the sidewall portion **12**, outside the carcass **3** in the tire width direction. The side rubber **6a** constitutes the outer surface of the sidewall portion **12** in the tire width direction. The side rubber **6a** is integrally formed with the tread rubber **5**.

(26) The inner liner **7** may be positioned on the tire inside of the carcass **3**, e.g., may be laminated on the tire inside of the carcass **3**. The inner liner **9** is made of butyl rubber with low air permeability, for example. The butyl rubber includes, for example, butyl rubber and its derivative, halogenated butyl rubber. The inner liner **9** is not limited to butyl rubber, but can be made of other rubber compositions, resins, or elastomers.

(27) Regarding the tire **1** according to this embodiment, although FIG. 2 illustrates only one side of the tire **1** with respect to the tire equatorial plane CL, the configuration of the tire **1** other than the conductive member **21** and the IC chip **22** (that is, when the portion of the tire **1** excluding the conductive member **21** and the IC chip **22** described below is referred to as the tire body portion, the configuration of the tire body portion) is symmetrical with respect to the tire equatorial plane CL. In other words, in the configuration of the tire **1**, the other side with respect to the tire equatorial plane of the tire is equivalent to the configuration illustrated in FIG. 2. However, the configuration of the tire **1** may be asymmetric with respect to the tire equatorial plane CL.

(28) In addition, the configuration of the tire **1** other than the conductive member **21** and the IC chip **22** described below may be different from those described above, as long as it does not depart from the purpose of the present disclosure.

(29) Next, the conductive member **21** and the IC chip **22**, embedded in the tire side portion **14** comprised of the sidewall portion **12** and the bead portion **13** in this embodiment, will be described.

(30) In this embodiment, the IC chip **22** (more specifically, at least the contact IC chip **22d**, described below) and the conductive member **21** can work together to configure a communication device **20** that enables communication with the outside world, with the conductive member **21** functioning as an antenna. In other words, when compared to the general RF tag (also commonly referred to as an RFID tag), in this embodiment, the IC chip **22** and the conductive member **21**

function as the IC chip of the RF tag and the antenna of the RF tag, respectively, so that the IC chip **22** and the conductive member **21** can work together to configure the communication device **20**.

(31) In this embodiment, the communication device **20**, comprising the IC chip **22** (more specifically, at least the contact IC chip **22d**, described below) and the conductive member **21**, is capable of wireless communication with a predetermined external device (e.g., a reader or reader/writer) located outside the tire **1**.

(32) The communication device **20** is suitably configured as a passive type, however, may also be configured as an active type.

(33) In this embodiment, as illustrated in FIGS. **1** and **2**, the conductive member **21** is embedded in the tire side portion **14** (more precisely, inside the tire side portion **14**) composed of the sidewall portion **12** and the bead portion **13**. More specifically, as illustrated in FIG. **2**, the conductive member **21** is embedded in the tire side portion **14**, outside the carcass **3** (more specifically, the ply body portion **3M** of the carcass ply **3a**) in the tire width direction. In this example, the conductive member **21** is embedded in the side rubber **6a** (i.e., inside the side rubber **6a**) forming the sidewall portion **12**.

(34) In the example illustrated in FIG. **2**, the conductive member **21** is embedded in the outside the carcass ply **3a** in the tire width direction so that at least part of the conductive member **21** in the tire radial direction contacts the carcass ply **3a** that constitutes the carcass **3**. However, in this example, the conductive member **21** may not be in contact with the carcass ply **3a**, as long as the entire conductive member **21** is positioned outside the carcass ply **3a** (and thus the carcass **3**) in the tire width direction.

(35) In this embodiment, the conductive member **21** is formed in the form of a strip, having a predetermined width (in FIGS. **1** and **2**, the width in the tire radial direction RD) and extending in the longitudinal direction (in FIG. **1**, the tire circumferential direction CD). The predetermined width may be constant or vary along the longitudinal direction.

(36) The conductive member **21** may, for example, be made of a plurality of conductive fibers in contact with each other to form a strip. In this case, compared to the case where the conductive member **21** is formed of a strip-shaped metallic plate that can function as an antenna, for example, the conductive member **21** has better communication properties, and also the conductive member **21** is more flexible and easier to be embedded inside the tire **1**. More specifically, the conductive member **21** may, for example, be made by laying a number of conductive fibers with the direction of extension approximately aligned as the longitudinal direction of the conductive member **21**, or randomly touching each other, to form a strip. However, the conductive member **21** is not limited to the above configuration, but may be formed, for example, of a strip-shaped metallic plate, etc. that can function as an antenna. The conductive fibers include, for example, steel fibers, stainless steel fibers, carbon fibers, conductive organic fibers such as conductive cotton fibers, etc.

(37) When the conductive member **21** is formed as a strip as described above, its cross-sectional shape in the width direction (equivalent to that of FIG. **2**) may be flat or slightly curved.

(38) The conductive member **21** may be embedded in the tire side portion **14** as its outer surface is covered with a coating material **23**. In this case, the conductive member **21** is easier to handle, making it easier to embed the conductive member **21** inside the tire side portion **14**. The coating material **23** includes, for example, coating rubber and coating resin.

(39) In this embodiment, the conductive member **21** embedded in the tire side portion **14** extends over at least partial area in the tire circumferential direction and at least partial area in the tire radial direction in the tire side portion **14**. In other words, the conductive member **21** has a predetermined width in the tire radial direction and extends with a predetermined length in the tire circumferential direction (see FIG. **1**).

(40) In the example illustrated in FIG. **1**, the conductive member **21** extends over an entire circumference in the tire circumferential direction in the tire side portion **14**. However, the conductive member **21** may not extend over the entire circumference in the tire circumferential

direction in the tire side portion **14**. For example, the conductive members **21** may extend only over a partial area in the tire circumferential direction in the tire side portion **14**, a plurality of divided conductive members **21** may extend intermittently along the tire circumferential direction, or each of the plurality of conductive members **21** extending only in a partial area in the tire circumferential direction may be arbitrarily embedded within the tire side portion **14** so that they do not contact each other. However, from the standpoint of reliability against failure of the communication device **20** of which the conductive member **21** is a part, and from the standpoint of the appearance of the tire side portion **14**, it is most preferable that the conductive member **21** extends over the entire circumference in the tire circumferential direction in the tire side portion **14**.

(41) The width of the conductive member **21** (or each conductive member **21** when comprising more than one) in the tire radial direction is not limited, but is preferably 10 to 40% of the tire cross-sectional height. If the width of the conductive member **21** in the tire radial direction is 10% or more of the tire cross-sectional height, the IC chip **22** can be brought into contact with the conductive member more easily, and if the width is 40% or less, the increase in tire weight can be properly controlled. From the same perspective, it is more preferable that the width of the conductive member **21** in the tire radial direction is 20 to 30% of the tire cross-sectional height.

(42) In this embodiment, the width of the conductive member **21** in the tire radial direction is constant along the tire circumferential direction (see FIG. 1), however it may not be constant.

(43) The length of the conductive member **21** (or each conductive member **21** when comprising more than one) along the tire circumferential direction is not particularly limited, but is preferably 90 degrees or more (i.e., $\frac{1}{4}$ or more of the entire tire circumference) as the central angle θ around the tire rotation axis O (see FIG. 1). If the length of the conductive member **21** along the tire circumferential direction is 90 degrees or more as the central angle θ , the IC chip **22** can be brought into contact with the conductive member more easily. From the same perspective, the length of the conductive member **21** along the tire circumferential direction is, as the central angle θ , more preferably 180 degrees or more, even more preferably 270 degrees or more, and most preferably 360 degrees (that is, the entire tire circumference) which is as illustrated in FIG. 1.

(44) The position of the conductive member **21** in the tire radial direction in the tire side portion **14** (i.e., inside the belt **4** in the tire radial direction) is not particularly limited, however the entire conductive member **21** is preferably embedded outside the bead core **2a** in the tire radial direction, more preferably embedded outside the outer edge **3e** in the tire radial direction of the ply turn-up portion **3T** in the tire radial direction.

(45) As illustrated in FIG. 2, in this example, the entire conductive member **21** is embedded at a location outside the outer edge **3e** in the tire radial direction of the ply turn-up portion **3T** in the tire radial direction and spanning over approximately the tire maximum width position.

(46) In this embodiment, as illustrated in FIGS. 1 and 2 (the enlarged view of part A), at least one (in this example, a plurality of) IC chip **22** is embedded in the tire side portion **14** (more precisely, inside the tire side portion **14**). More specifically, as illustrated in FIG. 2, at least one (in this example, a plurality of) IC chip **22** is embedded in the tire side portion **14**, outside the carcass **3** (more specifically, the ply body portion **3M** of the carcass ply **3a**) in the tire width direction. In this example, as illustrated in the enlarged view of part A in FIG. 2, at least one (in this example, a plurality of) IC chip **22** is embedded in a coating material **23** covering the conductive member **21**.

(47) As illustrated in FIGS. 1 and 2 (the enlarged view of part A), each IC chip **22** is preferably embedded in a position that approximately corresponds to the conductive member **21** in the tire circumferential direction and the tire radial direction.

(48) In this embodiment, at least one (in this example, a plurality) of the at least one IC chip **22** is in contact with the conductive member **21** without a mechanical connection, as illustrated in the enlarged view of part A in FIG. 2.

(49) In this document, the IC chip **22** “contacts (is in contact with) the conductive member **21** without a mechanical connection” means that the IC chip **22** does not adhere (is not adhered) (i.e.,

is not mechanically connected) to the conductive member **21** by, for example, solder, etc., however merely contacts (is in contact with) the conductive member **21**. Hereinafter, to contact (being in contact) without a mechanical connection is also referred to simply as “to contact (being in contact)” in this document.

(50) In this document, the IC chip **22** that contacts (or is in contact with) the conductive member **21** without a mechanical connection is referred to as the “contact IC chip **22c**”, and the IC chip **22** that does not contact (or is not in contact with) the conductive member **21** is referred to as the “non-contact IC chip **22d**”.

(51) In this embodiment, as described above, at least one of the at least one (in this example, a plurality of) IC chip **22** is a contact IC chip **22c**, which contacts the conductive member **21** without a mechanical connection.

(52) In this disclosure, the “IC chip **22**” refers to an IC chip **22** that does not have a separate antenna component functioning as an antenna within it.

(53) When the IC chip **22** is electrically coupled to the conductive member **21** as an antenna, for example, by contacting with the conductive member **21**, the IC chip **22** functions as a part of the communication device **20** comprised of both of them, and operates, for example, by dielectric-electromotive force generated by the radio waves received by the conductive member **21**. The IC chip **22**, for example, has a controller and a storage section.

(54) The storage section may store any information. For example, the storage section may store the identification information of the tire **1**. The identification information of the tire **1** is, for example, the unique identification information of the tire **1** that can identify each tire on a tire-by-tire basis, such as the manufacturer of the tire **1**, the manufacturing plant, the date of manufacture, and so on. The storage section may also store the history information of the tire such as the distance traveled, number of sudden braking, number of sudden starts, number of sudden turns, etc. In addition, for example, sensor(s) detecting tire internal temperature, tire internal pressure, tire acceleration, etc. may be provided in the tire lumen, and the storage section may store the detection information detected by the sensor(s). In this case, the communication device **20** comprised of the IC chip **22** and the conductive member **21** can wirelessly communicate with the sensor(s) through the conductive member **21** as an antenna to obtain the detection information from the sensor(s).

(55) The controller is configured, for example, to be able to read the information from the storage section.

(56) When the communication device **20** is composed of the IC chip **22** and the conductive member **21**, the communication device **20** is configured to receive, by the conductive member **21**, the information transmitted by radio waves or magnetic fields from a predetermined external device (e.g., a reader or reader/writer) located outside the tire **1**. Rectification (in the case of radio waves) or resonance (in the case of magnetic fields) generates electric power in the conductive member **21** of the communication device **20**, and the storage section and the controller of the IC chip **22** perform the prescribed operation. For example, the controller reads the information in the storage section and returns (transmits) it to the above prescribed external device from the conductive member **21** by radio waves or magnetic fields. The above prescribed external device receives radio waves or magnetic fields from the communication device **20**. The above prescribed external device can acquire the information stored in the storage section of the IC chip **22** of the communication device **20** by retrieving the received information.

(57) When the plurality of IC chips **22** are electrically coupled to the conductive member **21** and function as a part of the communication device **20** comprised of both of them, they are configured to transmit the same information to the above prescribed external device, for example.

(58) In this embodiment, as illustrated in the enlarged view of part A in FIG. 2, at least one (in this example, a plurality of) IC chip **22** is embedded in the coating material **23** covering the conductive member **21**, and at least one (in this example, more than one) of the at least one IC chip **22** is in contact with the conductive member **21**.

(59) In this case, for example, at least one IC chip **22** can be a contact IC chip **22c** that is in contact with the conductive member **21**, for example, by pre-coating the surface of the conductive member **21** with the coating material **23** in which at least one, preferably a plurality of IC chips **22** are embedded, and embedding the conductive member **21** coated with the coating material **23** inside the tire side portion **14**.

(60) As a method of embedding the conductive member **21** coated with the coating material **23** inside the tire side portion **14**, for example, the conductive member **21** coated with the coating material **23** may be placed on a raw tire during the manufacturing of the tire **1**, housed inside a tire molding die, and vulcanized.

(61) In this embodiment, although not illustrated in figures, at least one (preferably, a plurality of) IC chip **22** may be embedded in the side rubber **6a** forming the sidewall portion **12**, so that at least one (preferably, more than one) of the at least one IC chip **22** is in contact with the conductive member **21**.

(62) In this case, at least one IC chip **22** can be a contact IC chip **22c** that is in contact with the conductive member **21**, for example, by embedding the conductive member **21**, which is not coated with a coating material **23** or the like, as it is, adjacent to the side rubber **6a** in which at least one, preferably a plurality of IC chips **22** are pre-embedded during tire molding.

(63) As a method of embedding the conductive member **21** inside the tire side portion **14**, for example, the conductive member **21** may be placed on a raw tire containing the IC chip **22**, housed inside a tire molding die, and vulcanized.

(64) In this embodiment, as illustrated in FIG. **1**, the tire **1** comprises a plurality of the contact IC chips **22c**, and at least two of the plurality of the contact IC chips **22c** are embedded at positions spaced apart from each other in the tire circumferential direction.

(65) Further, in this embodiment, as illustrated in FIG. **1**, the tire **1** comprises a plurality of the contact IC chips **22c**, and at least two of the plurality of the contact IC chips **22c** are embedded at positions spaced apart from each other by 90 degrees or more as the central angle θ in the tire circumferential direction.

(66) Furthermore, in this embodiment, as illustrated in FIG. **1**, the tire **1** comprises a plurality of the contact IC chips **22c**, and at least two of the plurality of the contact IC chips **22c** are embedded at positions spaced apart from each other in the tire radial direction.

(67) More specifically, in the example in FIG. **1**, the tire **1** comprises ten IC chips **22**. Of the ten IC chips **22**, six are the contact IC chips **22c** (**22c1-22c6**) and four are the non-contact IC chips **22d** (**22d1-22d4**). All of the six contact IC chips **22c** (**22c1-22c6**) are embedded at positions spaced apart from each other in the tire circumferential direction. For example, the contact IC chips **22c1**, **22c3**, and **22c5** are embedded at positions spaced apart from each other by 90 degrees or more as the central angle θ in the tire circumferential direction. Further, for example, the contact IC chips **22c1** and **22c5** are embedded at positions spaced apart from each other in the tire radial direction.

(68) Thus, the fact that the tire **1** comprises the plurality of contact IC chips **22c** facilitates the continued operation of the communication device **20** comprised of the contact IC chips **22c** and the conductive member **21**, because even if at least some of them are damaged or failed, the other contact IC chips **22c** will function. In addition, the fact that at least two contact IC chips **22c** are embedded at positions spaced apart from each other in the tire circumferential direction or the tire radial direction, and in particular, sufficiently spaced apart from each other in the tire circumferential direction, reduces the possibility of damage or failure of both of the two contact IC chips **22c** due to the same circumstances, compared to the case where the two contact IC chips **22c** are embedded in the same position each other in the tire circumferential direction or the tire radial direction. Thus, the above configuration improves the reliability against failure of the communication device **20** comprised of the contact IC chips **22c** and the conductive member **21**.

(69) As in the example illustrated in FIG. **1**, when the tire **1** also comprises at least one non-contact IC chip **22d** (in the example of FIG. **1**, the non-contact IC chips **22d1-22d4**), at least some of them

may be capacitively coupled to the conductive member **21**, and as such, it is preferred that at least some of them are capacitively coupled to the conductive member **21**. In this case, the non-contact IC chip **22d**, which is capacitively coupled to the conductive member **21**, can also function as a part of the communication device **20**, thereby making the communication device **20** more reliable against failure.

(70) Here, the term “capacitively coupled” means that two conductive members are placed apart from each other and are electrically coupled by the capacitance between them.

(71) In this embodiment, from the viewpoint of ensuring the backup when at least some contact IC chips **22c** are damaged or failed, the number of contact IC chips **22c** embedded in the tire **1** (more specifically, inside the tire side portion **14**) is preferably three or more, and more preferably five or more. On the other hand, from the viewpoint of reducing cost increase, the number of contact IC chips **22c** embedded in the tire **1** (more specifically, inside the tire side portion **14**) is preferably fifteen or less, and more preferably ten or less.

(72) Here, considering the case where the IC chip **22** is embedded in the coating material **23** of the conductive member **21** or the IC chip **22** is embedded in the side rubber **6a**, for example, in order to obtain the tire **1** of this embodiment; if the number of IC chips **22** embedded is increased, even if the embedded positions of the IC chips **22** are not so precisely controlled (in other words, even if they are scattered and embedded roughly to some extent), the number of IC chips **22** that will be the contact IC chips **22c** can be increased.

(73) Next, the main effects of the above-mentioned embodiments are summarized below, again as necessary.

(74) First, in this embodiment, the tire comprises the conductive member **21** embedded in the tire side portion **14** comprised of the sidewall portion **12** and the bead portion **13**, outside the carcass **3** in the tire width direction, and extending over at least partial area in the tire circumferential direction and at least partial area in the tire radial direction in the tire side portion **14**; and at least one IC chip embedded in the tire side portion **14**, and at least one of the at least one IC chip **22** is a contact IC chip **22c** that contacts the conductive member **21** without a mechanical connection.

(75) This eliminates the need to mechanically connect the IC chip **22** and the antenna, or to cover them collectively with rubber or resin in advance for reinforcement, thus the communication device **20** can be simply configured with the IC chip **22** (more specifically, the contact IC chip **22c**, etc.) and the conductive member **21**. In addition, since the IC chip **22** is not mechanically connected to the conductive member **21**, the risk of disconnection failure, etc. of the connection is reduced and the durability of the communication device **20** is improved. That is, according to this embodiment, it is possible to improve the durability of the communication device **20** with a simple configuration.

(76) Further, according to this embodiment, since the conductive member **21**, which functions as an antenna, is embedded in the tire side portion **14**, where many metallic members, etc. which could be an obstacle to wireless communication are generally not located, the communication device **20** comprised of the IC chip **22** and the conductive member **21** can be excellent in communication.

(77) In this embodiment, the conductive member **21** may be made of a plurality of conductive fibers in contact with each other to form a strip.

(78) In this case, compared to the case where the conductive member **21** is formed of a strip-shaped metallic plate that can function as an antenna, for example, the conductive member **21** has better communication properties, and also the conductive member **21** is more flexible and easier to be embedded inside the tire **1**.

(79) In this embodiment, the conductive member **21** extends entirely in the tire circumferential direction in the tire side portion **14**.

(80) In this case, since the number of contact IC chips **22c** can be increased compared to the case where the conductive member **21** does not extend entirely in the tire circumferential direction in the tire side portion **14**, even if any one of the contact IC chips **22c** is damaged or failed, the other contact IC chips **22c** can continue to function as a part of the communication device **20** comprised

of the contact IC chips **22c** and the conductive member **21**, thus improving the reliability of the communication device **20** against failure. In addition, compared to the case where the conductive member **21** does not extend entirely in the circumferential direction in the tire side portion **14**, the possibility of localized bulging, etc. due to embedding of the conductive member **21** in the tire side portion **14** is reduced, and the appearance defects of the tire **1** can be suppressed. That is, according to this embodiment, the reliability of the communication device **20** against failure can be improved, and the appearance defects due to embedding of the communication device in the tire side portion **14** can be suppressed.

(81) In this embodiment, the tire **1** comprises a plurality of the contact IC chips **22c**, and at least two of the plurality of the contact IC chips **22c** are embedded at positions spaced apart from each other in the tire circumferential direction.

(82) Further, in this embodiment, the tire **1** comprises a plurality of the contact IC chips **22c**, and at least two of the plurality of the contact IC chips **22c** are embedded at positions spaced apart from each other by 90 degrees or more as the central angle θ in the tire circumferential direction.

(83) Furthermore, in this embodiment, the tire **1** comprises a plurality of the contact IC chips **22c**, and at least two of the plurality of the contact IC chips **22c** are embedded at positions spaced apart from each other in the tire radial direction.

(84) In these cases, as mentioned above, it is easier to continue to operate the communication device **20** comprised of the contact IC chips **22c** and the conductive member **21**, even if at least some of the contact IC chips **22c** are damaged or failed. In addition, the fact that at least two contact IC chips **22c** are embedded at positions spaced apart from each other in the tire circumferential direction or the tire radial direction reduces the possibility of damage or failure of both of the two contact IC chips **22c** due to the same circumstances. Thus, the reliability of the communication device **20** comprised of the contact IC chips **22c** and the conductive member **21** against failure is improved. For the same reason, the reliability of the communication device **20** against failure is further improved when at least two contact IC chips **22c** are embedded at positions spaced apart from each other by 90 degrees or more as the central angle θ in the tire circumferential direction.

(85) In this embodiment, the IC chip **22** is embedded in a coating material covering the conductive member **21**.

(86) In this case, by covering the surface of the conductive member **21** with the coating material **23** in which the IC chip **22** is embedded in advance, the IC chip **22** can be prepared at the same time as the conductive member **21** is prepared, and thus the tire **1** with the IC chip **22** can be efficiently manufactured.

(87) In this embodiment, the IC chip **22** is embedded in a side rubber **6a** forming the sidewall portion **12**.

(88) In this case, the IC chip **22** can only be embedded in the side rubber **6a** (raw rubber) in advance during the tire molding process, and thus the tire **1** with the IC chip **22** can be efficiently manufactured.

Second Embodiment

(89) Next, the tire **10** according to the second embodiment of this disclosure will be described with reference to FIG. 3.

(90) In FIG. 3, the components similar to those illustrated in FIGS. 1 and 2 are given the same reference numerals as in FIGS. 1 and 2, and their descriptions are omitted.

(91) The tire **10** according to the second embodiment of this disclosure differs from the tire **1** according to the first embodiment of this disclosure only in that it comprises a side-reinforcing rubber **6b** on the inside of the carcass **3** in the tire width direction in the sidewall portion **12**, and the conductive member **21** and the IC chip **22** are embedded in the side-reinforcing rubber **6b**, and the other points are the same as the tire **1** according to the first embodiment.

(92) As illustrated in FIG. 3, the tire **10** according to the second embodiment of this disclosure

comprises a nearly crescent-shaped side-reinforcing rubber **6b** inside the carcass **3** (more specifically, the ply body portion **3M** of the carcass ply **3a**) in the tire width direction in the sidewall portion **12**, between the carcass **3** and the inner liner **7**.

(93) In this embodiment, the conductive member **21** and the IC chip **22** are embedded inside the side-reinforcing rubber **6b**, more specifically, adjacent to the carcass ply **3a** in the example in FIG. **3**. In other words, in this embodiment, the conductive member **21** (and the IC chip **22**) is embedded inside the carcass **3** in the tire width direction in the tire side portion **14**.

(94) This reduces the risk of the communication device **20**, comprised of the IC chip **22** and the conductive member **21**, being damaged or malfunctioning due to foreign objects outside the tire, as compared to, for example, when the conductive member **21** (and IC chip **22**) is embedded outside the carcass **3** in the tire side portion **14** in the tire width direction.

(95) The other configuration and effects of the tire **10** according to this embodiment are the same as those of the tire **1** according to the first embodiment described above.

INDUSTRIAL APPLICABILITY

(96) The tire according to this disclosure can be suitably used for any type of pneumatic tire, for example, pneumatic tire for passenger cars, trucks, and buses, etc.

REFERENCE SIGNS LIST

(97) **1, 10** Tire **11** Tread portion **12** Sidewall portion **13** Bead portion **14** Tire side portion **2a** Bead core **2b** Bead filler **3** Carcass **3a** Carcass ply **3M** Ply body portion **3T** Ply turn-up portion **3e** Outer edge in the tire radial direction of ply turn-up portion **4** Belt **4a** Belt layer **5** Tread rubber **6a** Side rubber **6b** Side-reinforcing rubber **7** Inner liner **20** Communication device **21** Conductive member **22** IC chip **22c, 22c1-22c6** Contact IC chip **22d, 22dl-22d4** Non-contact IC chip **23** Coating material of conductive member **CL** Tire equatorial plane **WD** Tire width direction **RD** Tire radial direction **CD** Tire circumferential direction **O** Tire rotation axis θ Central angle

Claims

1. A tire having a carcass comprised of at least one carcass ply extending between a pair of bead portions through a pair of sidewall portions, wherein the tire comprises: a conductive member embedded in a tire side portion comprised of the sidewall portion and the bead portion, either outside or inside the carcass in the tire width direction, and extending over at least partial area in the tire circumferential direction and at least partial area in the tire radial direction in the tire side portion; and at least one IC chip embedded in the tire side portion, wherein at least one of the at least one IC chip is a contact IC chip that contacts the conductive member without a mechanical connection, the conductive member is made of a plurality of conductive fibers in contact with each other to form a strip, the tire comprises a plurality of the IC chips, and the plurality of the IC chips include the at least one contact IC chip and at least one non-contact IC chip that is free of contact with the conductive member and capacitively coupled to the conductive member.
2. The tire according to claim 1, wherein the conductive member extends around the entire circumference of the tire side portion.
3. The tire according to claim 2, comprising a plurality of the contact IC chips, and wherein at least two of the plurality of the contact IC chips are embedded at positions spaced apart from each other in the tire circumferential direction.
4. The tire according to claim 2, comprising a plurality of the contact IC chips, and wherein at least two of the plurality of the contact IC chips are embedded at positions spaced apart from each other in the tire radial direction.
5. The tire according to claim 2, wherein the IC chips are embedded in a side rubber forming the sidewall portion.
6. The tire according to claim 2, wherein the IC chips are embedded in a coating material covering the conductive member.

7. The tire according to claim 1, comprising a plurality of the contact IC chips, and wherein at least two of the plurality of the contact IC chips are embedded at positions spaced apart from each other in the tire circumferential direction.
 8. The tire according to claim 7, wherein the at least two of the plurality of the contact IC chips are embedded at positions spaced apart from each other by 90 degrees or more as the central angle in the tire circumferential direction.
 9. The tire according to claim 7, wherein the IC chips are embedded in a side rubber forming the sidewall portion.
 10. The tire according to claim 7, wherein the IC chips are embedded in a coating material covering the conductive member.
 11. The tire according to claim 1, comprising a plurality of the contact IC chips, and wherein at least two of the plurality of the contact IC chips are embedded at positions spaced apart from each other in the tire radial direction.
 12. The tire according to claim 1, wherein the IC chips are embedded in a side rubber forming the sidewall portion.
 13. The tire according to claim 1, wherein the IC chips are embedded in a coating material covering the conductive member.
 14. The tire according to claim 1, wherein the conductive member is made by laying the plurality of conductive fibers with a direction of extension approximately aligned as a longitudinal direction of the conductive member.
-